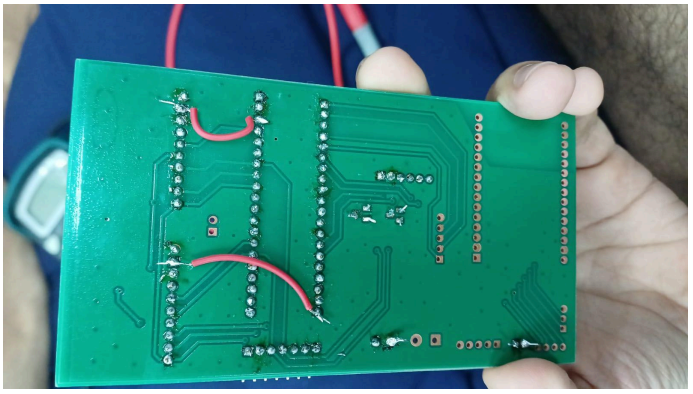
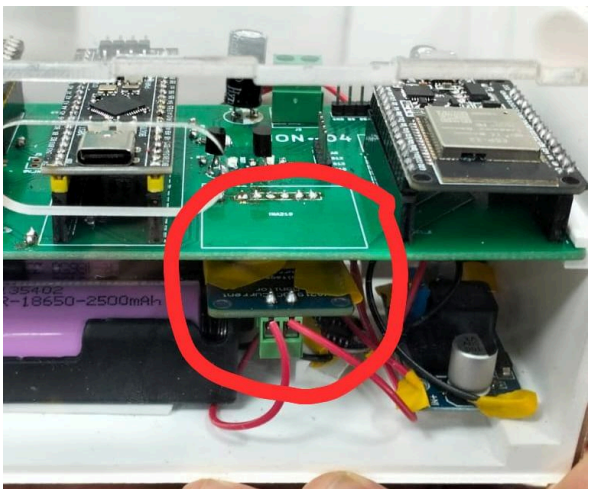
PCB known issues & workarounds:

1) We had been interfacing with the two Protocentral MAX30001 boards using an Arduino Mega board initially for code development. During that phase we had been powering the two MAX boards using 5V. Then PCB was sent for fabrication to the IITB PCB lab with the same connection that the two boards would be powered using 5V coming from the MCU. Our final goal was to use the STM32 blackpill so that we can run the DSP algorithms on-board. Although we knew that stm32 runs on 3V3 logic, we expected that using the 5V pin on the stm32 we can power both the boards. However when we tried it on the breadboard, both the MAX started giving noisy signals. When we replaced 5V by 3V3, the signal quality became just like what we were getting with the Arduino. But this finding came too late and afterwards we had to cut the 5V trace and connect 3V3 from the stm32 using wires.



2) Looking down on the top layer, the pin sequence for the INA219 board was inverted. We had to attach the INA219 from the bottom side, taking advantage of the through-hole nature of the solder pads.



3) For UART between the STM32 blackpill and the ESP32, we mistakenly connected the TX (stm32) with TX (esp32), similarly RX with RX. Whereas, it should have been the opposite. Again we had to cut the traces and make the opposite connections using wires.

